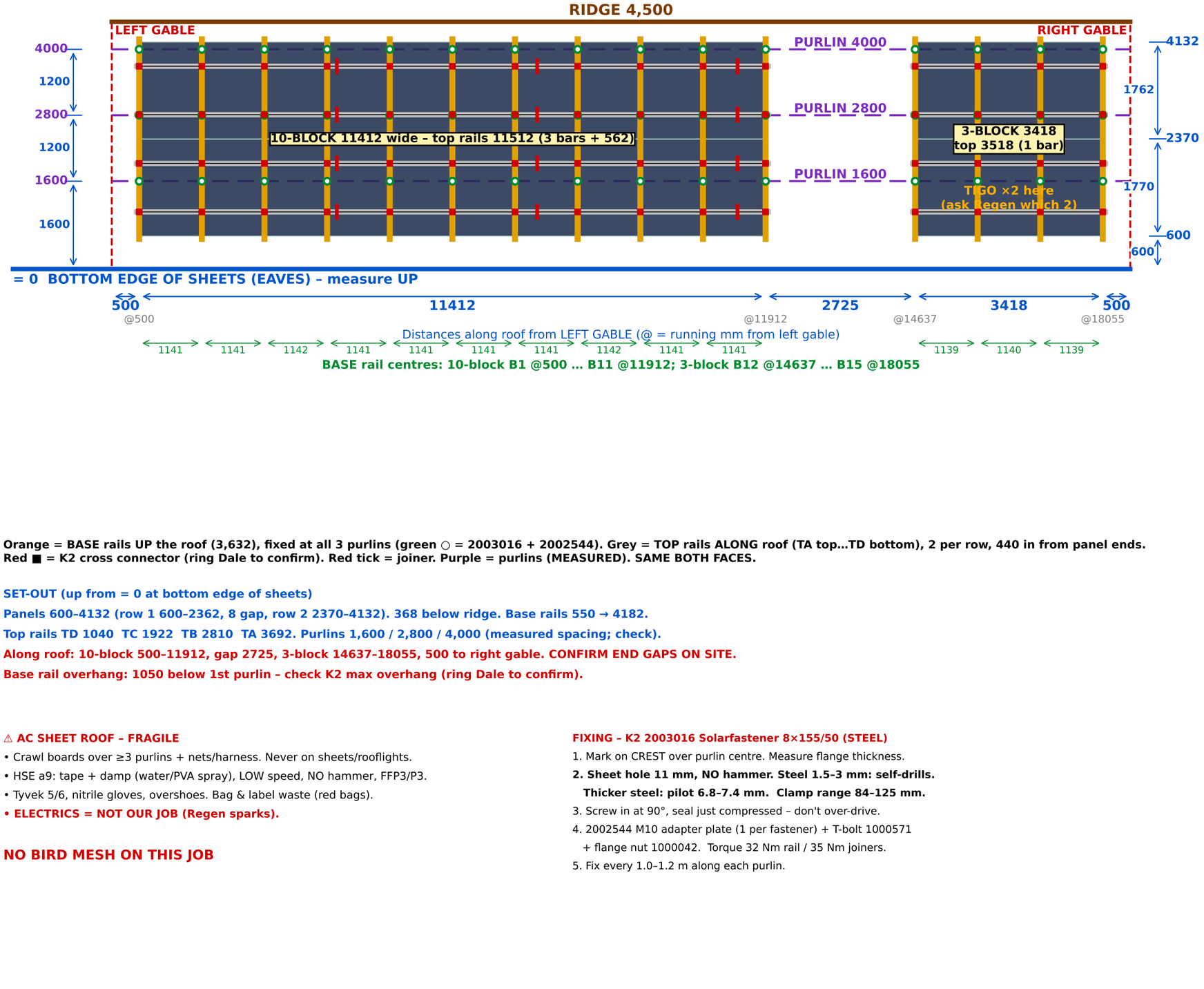
FINAL v8 – GREENACRE BARN – 26 PANELS PER FACE – PORTRAIT ON OVERLAY RAILS

LEWIS – Greenacre Farm, Murcott OX5 2RE | Regen – Dale Collins | BARN (52 = 26 per face) – AC corrugated sheet on STEEL purlins | Install Wed 4 Nov 2026 | v8

ONE FACE SHOWN – BOTH FACES IDENTICAL (confirm mirror / same end on other face) | 2×10 block + 2×3 block | 52 total



Orange = BASE rails UP the roof (3,632), fixed at all 3 purlins (green ○ = 2003016 + 2002544). Grey = TOP rails ALONG roof (TA top…TD bottom), 2 per row, 440 in from panel ends.
Red ■ = K2 cross connector (ring Dale to confirm). Red tick = joiner. Purple = purlins (MEASURED). SAME BOTH FACES.

SET-OUT (up from = 0 at bottom edge of sheets)

Panels 600–4132 (row 1 600–2362, 8 gap, row 2 2370–4132). 368 below ridge. Base rails 550 → 4182.

Top rails TD 1040 TC 1922 TB 2810 TA 3692. Purlins 1,600 / 2,800 / 4,000 (measured spacing; check).

Along roof: 10-block 500–11912, gap 2725, 3-block 14637–18055, 500 to right gable. CONFIRM END GAPS ON SITE.

Base rail overhang: 1050 below 1st purlin – check K2 max overhang (ring Dale to confirm).

⚠ AC SHEET ROOF – FRAGILE

- Crawl boards over ≥3 purlins + nets/harness. Never on sheets/rooflights.
- HSE a9: tape + damp (water/PVA spray), LOW speed, NO hammer, FFP3/P3.
- Tyvek 5/6, nitrile gloves, overshoes. Bag & label waste (red bags).
- **ELECTRICS = NOT OUR JOB (Regen sparks).**

NO BIRD MESH ON THIS JOB

FIXING – K2 2003016 Solarfastener 8×155/50 (STEEL)

1. Mark on CREST over purlin centre. Measure flange thickness.
2. **Sheet hole 11 mm, NO hammer. Steel 1.5–3 mm: self-drills.**
   Thicker steel: pilot 6.8–7.4 mm. Clamp range 84–125 mm.
3. Screw in at 90°, seal just compressed – don't over-drive.
4. 2002544 M10 adapter plate (1 per fastener) + T-bolt 1000571
   + flange nut 1000042. Torque 32 Nm rail / 35 Nm joiners.
5. Fix every 1.0–1.2 m along each purlin.

PER FACE

BASE: 15 × 3,632 (10-block 11, 3-block 4) ~1,140 apart

TOP: 4 × 11512 (3650+3650+3650+562) + 4 × 3518

Fixings 45 Cross conn. 60 End clamps 16 Mid 44 Joiners 12

BOTH FACES – see load list sheet

Fixings 90 • cross connectors 120 • panels 52

OPTIMISERS: 2 × Tigo TS4-A-O on the 3-block

(ask Regen which 2 panels) – see Tigo sheet

TOOLS – tick when loaded

☐ Drill/driver + spare batts (NO hammer) ☐ 11 mm + 7 mm bits
☐ Sockets 13/15/17 + T30 ☐ Torque wrench 32/35 Nm
☐ Hacksaw + 24/32 TPI blades (alu), file/deburr
☐ Tape, chalk line, marker, masking tape, square
☐ WIRE CUTTERS ☐ STANLEY KNIFE ☐ Cable ties
☐ Sealant gun + sealant
☐ JSP Force 8 P3 / FFP3, Tyvek, gloves, overshoes
☐ Spray bottle water+PVA, red asbestos bags, labels, OB1 tape, wipes
☐ Crawl boards, harness/lanyards, edge protection
☐ **NO electrical tools – electrics are Regen's**

FLAGS

1. Purlins measured – fix every base rail at all 3.
2. Confirm end gaps (500 each gable) + mirror/same end on face 2.
3. Rails 27 SHORT, plates 5 SHORT, cross connectors ? – ring Dale to confirm.
4. Flange > 3 mm → pilot 6.8–7.4 mm.

Not for upload – John's info only. Not a structural design.

FINAL LOAD LIST + DELIVERED – GREENACRE BARN – 52 PANELS (26 per face)

LEWIS – Greenacre Farm, Murcott OX5 2RE | Regen – Dale Collins | BARN (52 = 26 per face) – AC corrugated sheet on STEEL purlins | Install Wed 4 Nov 2026 | v8

Both faces: base rails UP roof fixed at 3 measured purlins, top rails ALONG roof on K2 cross connectors

Item	NEED (52)	DELIVERED (+/–)
JA JAM54D40-460/LB panels	**52**	**not stated**
K2 rail 3.65 m bars	**64**	**37 (-27)**
Rail joiners (sleeves)	**24**	**24 (+0)**
End clamps 30 mm	**32**	**40 (+8)**
Mid clamps 8 mm	**88**	**100 (+12)**
K2 cross connectors (ring Dale to confirm)	**120**	**36 plates?**
2003016 Solarfastener 8×155/50	**90**	**120 (+30)**
2002544 M10 adapter plates	**90**	**85 (-5)**
M10 T-bolt 1000571	**90**	**200 (+110)**
Flange nut 1000042	**90**	**200 (+110)**
Tigo TS4-A-O optimisers	**2**	**2 (+0)**
Black plastic clips (end caps?)	**?**	**? qty**

TOOLS – tick when loaded

☐ Drill/driver + spare batts (NO hammer) ☐ 11 mm + 7 mm bits

☐ Sockets 13/15/17 + T30 ☐ Torque wrench 32/35 Nm

☐ Hacksaw + 24/32 TPI blades (alu), file/deburr

☐ Tape, chalk line, marker, masking tape, square

☐ WIRE CUTTERS ☐ STANLEY KNIFE ☐ Cable ties

☐ Sealant gun + sealant

☐ JSP Force 8 P3 / FFP3, Tyvek, gloves, overshoes

☐ Spray bottle water+PVA, red asbestos bags, labels, OB1 tape, wipes

☐ Crawl boards, harness/lanyards, edge protection

☐ **NO electrical tools – electrics are Regen's**

⚠ AC SHEET ROOF – FRAGILE

- Crawl boards over ≥3 purlins + nets/harness. Never on sheets/rooflights.
- HSE a9: tape + damp (water/PVA spray), LOW speed, NO hammer, FFP3/P3.
- Tyvek 5/6, nitrile gloves, overshoes. Bag & label waste (red bags).
- **ELECTRICS = NOT OUR JOB (Regen sparks).**

SHORTFALLS vs ON-SITE DELIVERY – ring Dale to confirm

- **Rails: need 64 × 3.65 m bars, have 37 → 27 SHORT**
- **Adapter plates 2002544: need 90, have 85 → 5 SHORT**
- **Cross connectors: need 120. '36 metal plates' – CONFIRM: cross connectors or adapter plates? If cross connectors → 84 SHORT**
- **Black plastic joining clips – qty ? (likely rail end caps) – check count**

OK: end clamps 40 (+8), mid 100 (+12), joiners 24 (+0), fasteners 120 (+30), T-bolts/nuts 200 (+110), Tigo 2 (0)

NO BIRD MESH ON THIS JOB

Rails: base 30 × 3632 (1 bar each); 10-block top 8 × 11512 = 24 full + 8 × 562 cut (2 bars); 3-block top 8 × 3518 (8 bars)

Cut-bar plan for 562s: 562+562+562+562+562+562 | 562+562

OPTIMISER FITTING – 2 × TIGO TS4-A-O – 12-panel OUTBUILDING

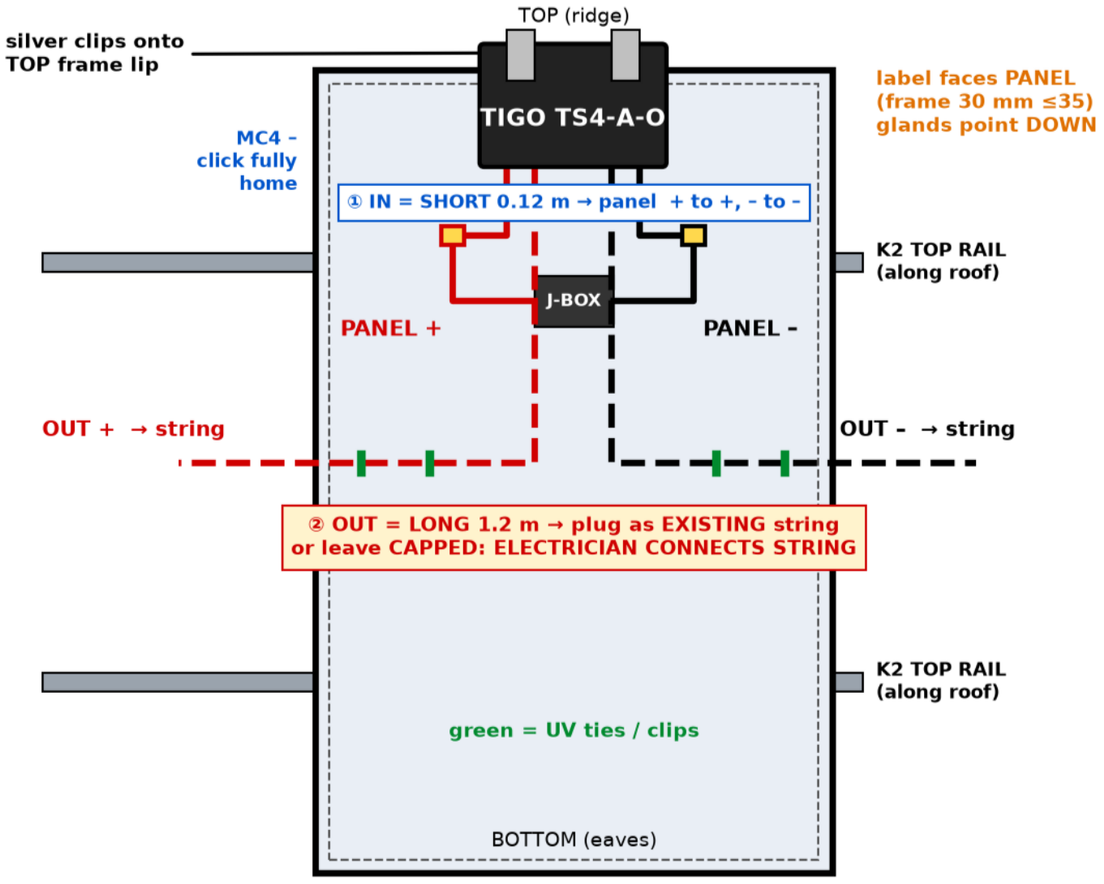
LEWIS – Greenacre Farm, Murcott OX5 2RE | Regen – Dale Collins | JA JAM54D40-460/LB PORTRAIT 1762×1134×30 on K2 overlay rails | v2

PHYSICAL FITTING + PLUG-IN ONLY. No string design, no inverter wiring – ELECTRICIAN CONNECTS THE STRING.

Elizabeth John Designs Ltd – installation sheet
Source: Tigo TS4-A Installation Manual v2.3 (002-00129-00) + TS4-A-O datasheet, tigoenergy.com

REAR VIEW – underside of one portrait panel

(looking at the back, top of panel = ridge end)



Cables tied to rail/frame – NEVER lying on the roof sheet.
No tension, no tight bends, drip loop at each plug.

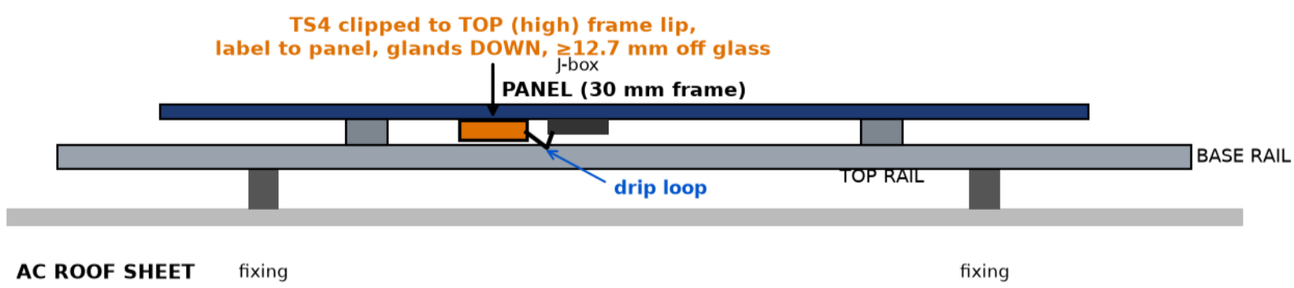


John's photo – Tigo TS4-A-O on site:
- **black box, yellow TS4-A-O label**
- **2 silver frame clips (top edge)**
- **4 MC4 leads: 2 short IN, 2 long OUT**

Guide: tigoenergy.com/documents/installation-manual-ts4-cca-tap

SIDE VIEW – across the slope

TS4, cables, glands & plugs must NOT touch the roof sheet (Tigo).
Keep airflow round it – never sit it in a valley/trough.



TIGO TS4-A-O – KEY FACTS

- Clips onto the TOP (high) edge of the panel frame – silver spring clips
- **JA frame is 30 mm (≤35 mm) → LABEL FACES THE PANEL, glands DOWN**
- No torque on clips. (Frameless only: M8 bolts to rail, 10.2 Nm – not us)
- IN 0.12 m short / OUT 1.2 m long, MC4
- **Ask Regen WHICH 2 panels**

⚠ SAFETY – TIGO WARNINGS

- **Ships switched ON – treat as LIVE**
- **NEVER connect/disconnect under load**
- SHORT IN leads to panel FIRST, then OUT – wrong order can damage it
- Don't mix connector brands / cut MC4
- Plugs out of rain; no jewellery
- Removing: OUT off string first
- Anything odd → STOP, ring sparky

STEPS (from Tigo's manual)

1. Ask Regen which 2 panels. Mark them on the layout sheet.
2. Peel the QR/barcode serial sticker off each TS4 and stick it on the layout sheet at that panel's spot (photo it too).
3. Clip TS4 onto the TOP edge of the panel frame, label facing the panel, cable glands DOWN, before the panel is clamped. Clear of the roof sheet.
4. FIRST plug the SHORT IN leads into the panel: + to +, – to –. Click, tug.
5. THEN the LONG OUT leads: plug on as the existing string, or cap & leave – ELECTRICIAN CONNECTS THE STRING.
6. Tie cables to rail/frame (UV ties), drip loops, no tension, nothing on the sheet. Then clamp the panel down.